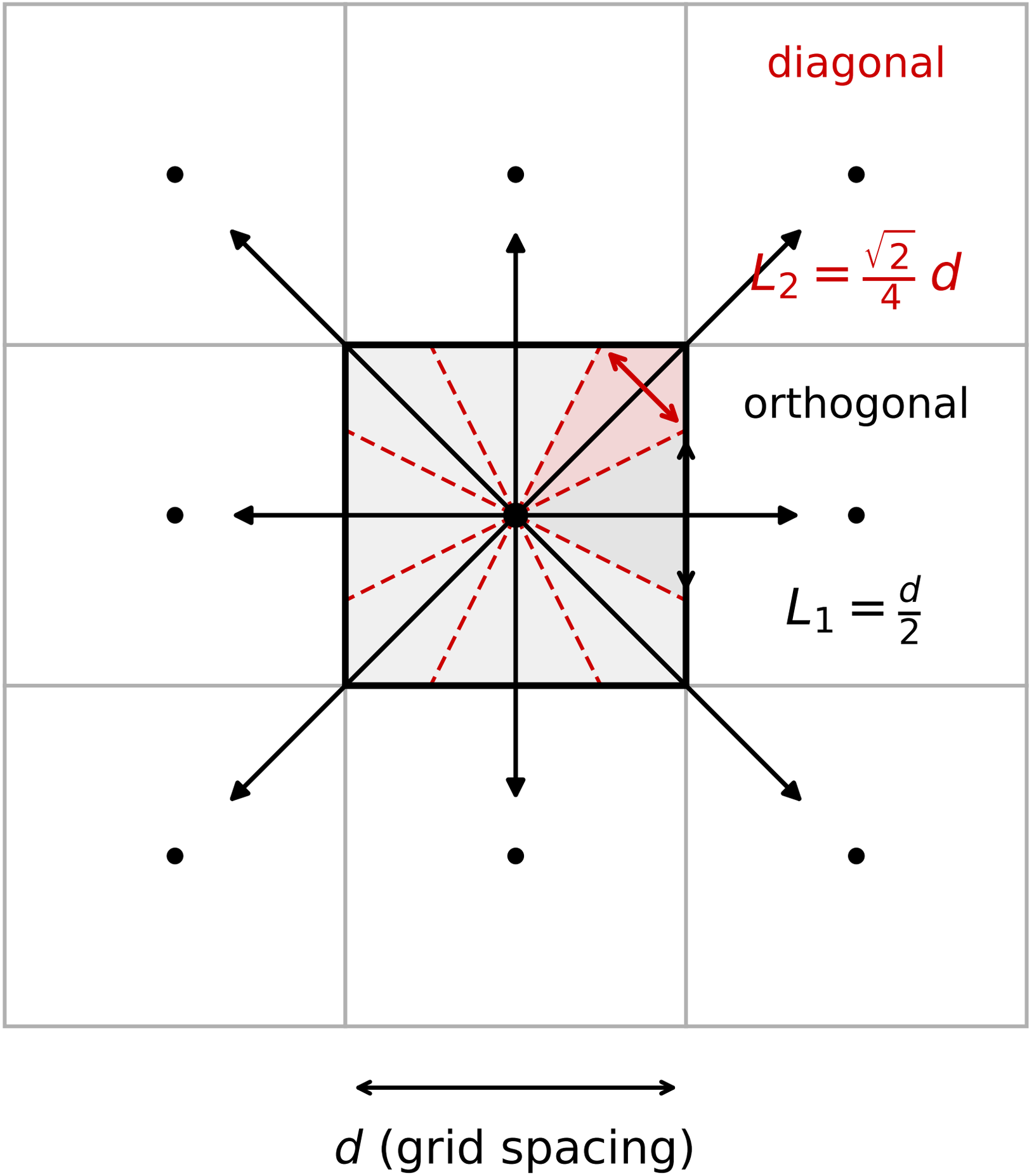
diagonal
$L_2 = \frac{\sqrt{2}}{4}\, d$
orthogonal
$L_1 = \frac{d}{2}$
d (grid spacing)